

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

PropSight: A Financial Monitoring and Reporting System for Data-Driven Rental Property Management

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The rental property business has long been recognized as a stable and accessible source of income for many individuals and families. Property owners who manage apartments, boarding houses, and small commercial spaces rely on rental income to support their daily needs, sustain investments, and build long-term financial security. In the Philippines, this sector plays a particularly important role in the housing ecosystem, as it provides accommodation for a growing number of people who are unable to afford homeownership due to rising property prices and economic constraints.

Recent trends indicate that renting is becoming more common among Filipino households. According to the Bangko Sentral ng Pilipinas (BSP) Consumer Finance Survey, approximately **11.3% of households nationwide were renting as of 2021**, an increase from 10.2% in 2018. This trend is even more pronounced in urban areas such as Metro Manila, where **34.9% of households are renters**, largely due to the high cost of purchasing property and slower income growth (Bangko Sentral ng Pilipinas [BSP], 2021; *BusinessWorld*, 2025). These figures highlight the growing dependence on rental housing, particularly in densely populated and economically active regions.

The increasing demand for rental housing is further driven by rapid urbanization and population growth. The Philippines has a population exceeding 112 million people, contributing to high population density in major urban centers and increased pressure on housing availability (Statista, 2024). Rental housing has therefore become a critical solution in accommodating migrants, students, workers, and low- to middle-income families. Studies also emphasize that rental housing plays a vital role in absorbing urban population growth, particularly in cities where land for residential development is limited (Philippine Institute for Development Studies [PIDS], 2018).

Despite its importance, the management of rental properties in the Philippines remains largely informal, especially among small to medium-scale landlords. Many property owners still rely on traditional methods such as handwritten notebooks, paper receipts, or basic spreadsheet tools to manage their finances. While these methods may be sufficient for a small number of tenants, they become increasingly inefficient as the scale of operations grows. Errors such as miscalculations, missing records, and inconsistent data entries are common, making it difficult to maintain accurate financial tracking.

This challenge is particularly evident among small-scale landlords, who play a crucial role in providing affordable housing but often face operational and financial constraints. Research highlights that these landlords significantly contribute to the supply of rental housing, yet they encounter limitations that hinder efficient property management and expansion (Abdullah et al., 2022). Unlike large property management companies, small-scale landlords typically manage their properties independently, handling tasks such as rent collection, expense monitoring, tenant management, and maintenance coordination without the support of structured systems.

In addition to operational challenges, the Philippine rental market presents complex financial dynamics. As of recent reports, the average rental yield in the country is approximately **5.11%**, indicating that rental properties remain a viable source of income for investors (Global Property Guide, 2024). However, profitability is influenced by factors such as vacancy rates, maintenance costs, and fluctuating rental prices. Some areas in Metro Manila experience vacancy rates ranging from **24% to 26%**, with certain locations exceeding 50%, reflecting an imbalance between supply and demand (Bamboo Routes, 2024). Meanwhile, rental prices continue to increase modestly, with an annual growth rate of approximately **2–3%** (Global Property Guide, 2024).

These conditions create a challenging environment for property owners, who must carefully monitor income, expenses, and occupancy rates to maintain profitability. However, without a centralized and organized system, it becomes difficult to gain a clear understanding of financial performance. When financial data is scattered across multiple notebooks, files, or spreadsheets, generating accurate reports and identifying trends—such as late payments, increasing maintenance costs, or declining occupancy rates—requires significant time and effort.

Furthermore, the lack of transparency and data integration limits the ability of property owners to make informed decisions. Financial planning, pricing strategies, and investment decisions are

often based on incomplete or outdated information. This not only affects short-term operations but also impacts long-term sustainability and growth. As rental businesses expand, the need for a more structured, efficient, and data-driven approach becomes increasingly critical.

The integration of digital technologies into property management offers a practical solution to these challenges. By automating financial tracking, consolidating records, and providing real-time insights, digital systems can significantly improve efficiency and accuracy. A centralized platform allows property owners to easily monitor income and expenses, generate financial reports, and analyze trends, enabling better decision-making and improved financial control.

In response to these challenges, this study proposes **PropSight: A Financial Monitoring and Reporting System for Data-Driven Rental Property Management**. The system is designed to address the limitations of manual financial management by providing a centralized, user-friendly platform for recording and monitoring financial transactions. It aims to automate key processes such as rent tracking, expense recording, and report generation, while also offering data-driven insights to support decision-making.

Through the implementation of automated reporting and financial analysis tools, PropSight seeks to enhance transparency, reduce human error, and improve operational efficiency among property owners. By empowering landlords with accurate and timely information, the system enables more informed financial planning and strategic decision-making. As the demand for rental housing continues to grow in the Philippines, the adoption of such technological solutions becomes essential in ensuring sustainable, efficient, and competitive property management practices.

1.2 Statement of the Problem

Despite the increasing number of rental properties and the growing dependence on rental income, many property owners continue to face difficulties in effectively managing and monitoring their financial transactions. These challenges often result in inefficiencies, inaccuracies, and limited financial insight. Many property owners still rely on manual methods such as notebooks, spreadsheets, or unorganized digital files, which can lead to errors, duplication of entries, and even data loss. Tracking rental payments, recording expenses, and

monitoring outstanding balances can become complicated, especially when managing multiple properties or tenants at the same time. Inconsistent record-keeping may also result in missed payments, overlooked expenses, or incorrect financial reporting, which can affect the overall financial condition of the property owner.

Furthermore, the absence of an integrated and centralized system makes it difficult for property owners to access real-time financial information whenever needed. This limits their ability to make timely and well-informed decisions regarding their investments and operations. Without proper tools, analyzing financial performance, identifying income trends, and detecting unnecessary expenses becomes challenging and time-consuming. Property owners may struggle to evaluate profitability, manage cash flow efficiently, and prepare accurate financial reports. In addition, the lack of automation increases the workload and the possibility of human error in handling financial data. Communication gaps between property owners and tenants regarding payments may also arise due to poor tracking systems. These issues can negatively affect overall property management, decision-making, and long-term financial stability. Therefore, there is a strong need for a reliable, accurate, and user-friendly system that can simplify financial tracking, improve data accuracy, provide real-time monitoring, and support better decision-making for property owners.

Specifically, this study aims to address the following problems:

1. Financial records of rental income and expenses are frequently managed manually or through fragmented systems, leading to disorganized data handling. This makes it difficult to maintain complete and reliable records, especially when dealing with multiple tenants and properties. Many property owners still rely on spreadsheets, notebooks, or separate applications to manage their financial data. This scattered approach often results in missing or duplicated entries. Over time, it becomes harder to track historical transactions accurately. Important financial information may be stored in different locations, making retrieval time-consuming. This lack of organization increases the risk of confusion in financial reporting. It also makes it challenging to monitor overall property performance effectively. Without a unified system, consistency in record-keeping is difficult to achieve. As a result, financial management becomes inefficient and prone to errors.

2. Property owners encounter difficulties in generating accurate, consistent, and timely financial reports. As a result, decision-making is often delayed or based on incomplete or outdated information. Preparing reports manually can take a significant amount of time and effort. This delay prevents property owners from quickly understanding their financial status. In many cases, reports may contain errors due to manual calculations. Inconsistent data sources can also affect the reliability of financial summaries. Property owners may struggle to compare income and expenses across different time periods. This makes it difficult to evaluate profitability accurately. Without timely reports, important financial decisions may be postponed. Outdated information can lead to poor planning and missed opportunities. Therefore, efficient reporting tools are essential for better financial management.

3. There is a lack of real-time monitoring of rental payments, outstanding balances, and operational expenses. This limits the ability of property owners to promptly track financial activities and respond to issues such as late payments. Many current systems only provide updates after manual entry or periodic checks. This delay prevents immediate awareness of financial changes. Property owners may not know right away if tenants fail to pay on time. Late detection of payment issues can affect cash flow and budgeting. Operational expenses may also go unnoticed until reviewed later. Without real-time updates, financial tracking becomes less reliable. It becomes harder to manage multiple tenants efficiently. Immediate action on financial issues is often not possible. This highlights the need for an automated monitoring system.

4. Financial data is not presented in a structured and analytical format, limiting its usefulness for informed and data-driven decision-making. Consequently, property owners may struggle to identify financial trends and optimize their management strategies. Raw financial data is often difficult to interpret without proper organization. Many systems fail to convert data into meaningful insights. Property owners may find it challenging to understand patterns in income and expenses. Without proper analysis tools, decision-making becomes based on assumptions. This can lead to ineffective financial strategies. Identifying profitable and unprofitable properties becomes harder. Trends such as seasonal income changes may go unnoticed. Opportunities for cost reduction may also be missed. Structured data presentation is essential for better understanding. Analytical tools help improve long-term financial planning.

5. Manual processes increase the risk of human errors, data inconsistencies, and reduced transparency in financial management. This can lead to discrepancies in records and decreased trust in the accuracy of financial information. When data is entered manually, mistakes are more likely to occur. Simple errors in numbers can affect overall financial reports. Inconsistencies may arise when different methods are used for recording data. These issues make it difficult to verify financial accuracy. Transparency is reduced when records are unclear or incomplete. Property owners may find it hard to justify financial results. Tenants and stakeholders may also lose trust in the system. Reconciliation of financial records becomes time-consuming. Errors may accumulate over time if not corrected. Therefore, automated and structured systems are needed to improve accuracy and trust.

1.3 Objectives of the Study

1.3.1 General Objective

Specifically, the study aims to: The primary objective of this study is to develop **PropSight: A Financial Monitoring and Reporting System for Data-Driven Rental Property Management** that will improve the efficiency, accuracy, and accessibility of financial management for rental property owners.

1.3.2 Specific Objectives

- 1. Develop** a centralized system that enables rental property owners to efficiently record and manage financial data, including rental income and operational expenses.
 - 2. Design and implement** a feature that facilitates the generation of accurate and timely financial reports for effective monitoring of property performance.
 - 3. Provide** a real-time monitoring mechanism for tracking rental payments, outstanding balances, and operational costs.
 - 4. Organize and present** financial data in a structured and analytical format to support data-driven decision-making.
 - 4. Minimize** errors, inconsistencies, and lack of transparency associated with manual financial management processes through system automation.
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1.4 Scope and Delimitations

Scope of the Study

This study focuses on the design and development of **PropSight: A Financial Monitoring and Reporting System for Data-Driven Rental Property Management**, a web-based application intended to assist rental property owners in managing their financial transactions in a more organized, efficient, and accurate manner.

The system is specifically designed for small- to medium-scale property owners such as those managing apartments, boarding houses, dormitories, and small commercial rental spaces. It aims to centralize financial records and improve monitoring of rental-related income and expenses.

The scope of the system includes the following features:

- Web-based recording and management of rental income and operational expenses
- Monitoring of tenant payments, due dates, and outstanding balances
- Automated generation of financial reports such as income summaries, expense breakdowns, and overall financial performance reports
- Real-time viewing of financial transactions and account status
- Organized presentation of financial data for easier analysis and interpretation
- User authentication for secure access of property owner accounts
- A responsive and user-friendly web interface accessible through modern web browsers

The system will be developed using web technologies and will utilize a centralized database to ensure data consistency, accessibility, and efficient management of financial records. It is intended to support property owners in improving financial transparency and enabling data-driven decision-making.

Delimitations of the Study

Despite its intended functionality and benefits, the system has the following limitations:

- The system is limited to financial monitoring and reporting and does not include full property management features such as tenant communication, messaging systems, or lease contract generation.

- It is dependent on user input; therefore, the accuracy and reliability of financial records depend on the correctness of the data entered by the property owner or user.
 - The system is designed primarily for small- to medium-scale rental property operations and may not fully support large-scale enterprise-level property management systems.
 - As a web-based application, the system requires an internet connection to access and use its features, which may affect usability in areas with poor or unstable connectivity.
 - The system does not include advanced technologies such as artificial intelligence-based financial forecasting or predictive analytics.
 - Security features are limited to basic authentication and data protection mechanisms and may not meet advanced enterprise cybersecurity standards.
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1.5 Significance of the Study

The proposed system, PropSight: A Financial Monitoring and Reporting System for Data-Driven Rental Property Management, is expected to provide benefits to the following individuals and groups:

Property Owners

The primary beneficiaries of this study are property owners who manage rental spaces such as apartments, boarding houses, dormitories, and small commercial units. The system will assist them in efficiently monitoring rental income, tracking operational expenses, and managing outstanding payments in a centralized platform. By replacing manual and fragmented record-keeping methods, PropSight reduces the likelihood of human errors, data inconsistencies, and missing records. It also saves time in generating financial reports and provides accurate, organized, and real-time financial information. As a result, property owners can make more informed and data-driven financial decisions, improving overall business efficiency and profitability.

Tenants

Although the system is primarily designed for financial monitoring from the property owner's perspective, tenants will also indirectly benefit from its implementation. With improved tracking and organized record-keeping of payments, tenants may experience more transparent and accurate billing systems. This reduces misunderstandings or disputes regarding rental

balances, due dates, and payment history. Ultimately, the system promotes clearer communication and accountability between tenants and property owners.

Future Researchers

This study can serve as a valuable reference for future researchers who intend to explore topics related to financial monitoring systems, rental property management systems, and data-driven decision-making tools. It may provide insights into system design, development methodologies, and functional requirements that can be adapted or enhanced in future studies. Additionally, it can serve as a foundation for improving or expanding similar systems in the context of small- to medium-scale business operations.

Developers

This study offers a framework for designing and implementing financial monitoring systems. It can serve as a guide for developers who intend to create similar web-based property management applications. Moreover, it demonstrates practical approaches to integrating financial tracking, reporting, and data visualization features. Developers can also gain insights into improving system usability, scalability, and functionality in real-world applications.

1.6 Definition of Terms

To ensure clarity and better understanding of the study, the following terms are defined as used in the context of this research:

PropSight - Is a proposed system designed to assist property owners in efficiently tracking and managing their rental income and expenses. It provides a centralized platform where all financial data related to rental properties can be recorded and accessed. The system is intended to simplify complex financial processes by organizing information in a structured manner. It allows users to monitor income, expenses, and overall financial performance in one place. Through its user-friendly interface, property owners can easily navigate and manage their records. The system also integrates real-time monitoring to ensure that all transactions are updated instantly. By automating financial tracking, it minimizes errors and reduces manual work. Overall, PropSight enhances efficiency and supports better financial decision-making for property owners.

Rental Property - A rental property refers to a residential or commercial space that is leased to tenants in exchange for regular payments. These properties may include houses, apartments, condominiums, or commercial establishments such as offices and retail stores. Property owners generate income through rent collected from tenants occupying these spaces. Managing rental properties involves tasks such as maintenance, rent collection, and tenant communication. It also requires monitoring occupancy and ensuring that lease agreements are followed. Proper management helps maintain the value and condition of the property. Rental properties serve as a stable source of income for many investors. They also play a significant role in providing housing and business spaces within a community.

Property Owner / Landlord - An individual or entity who owns and manages rental properties. They are responsible for overseeing tenants, collecting rent, monitoring expenses, handling bookings, and managing maintenance-related concerns. They may also set rental policies, screen potential tenants, and ensure that the property complies with local laws and regulations. Effective landlords maintain clear communication with tenants to address concerns promptly and build a positive rental experience. They are also responsible for maintaining proper documentation such as lease agreements, receipts, and financial records. In some cases, landlords may hire property managers to assist with daily operations. They must also ensure that safety standards are met within the property.

Tenant - A person who rents a property from the owner under agreed terms and conditions, including rental fees, duration of stay, and payment schedules. Tenants are expected to follow the rules stated in the lease agreement and take reasonable care of the property. They also have the right to a safe and habitable living environment. Good tenants communicate issues early and pay rent on time. They should respect neighbors and avoid causing disturbances within the property. Tenants are also responsible for reporting damages or issues as soon as possible. In some agreements, they may share responsibilities such as minor maintenance or utility payments.

Booking System - A system feature that allows tenants to reserve or schedule rental units or spaces in advance. It helps property owners manage availability, reservations, and occupancy efficiently. The system can include calendars, automated confirmations, and reminders to avoid scheduling conflicts. It may also allow online payments and provide real-time updates for both tenants and landlords. Advanced booking systems may include user accounts and booking histories. They can generate reports for occupancy rates and revenue tracking. Some systems also integrate notifications through email or mobile apps for better communication.

Maintenance Request - A system feature that allows tenants or users to report issues, damages, or repair needs within the rental property. It helps property owners track and respond to maintenance concerns efficiently. Requests can be prioritized based on urgency and assigned to maintenance staff. This feature often includes status updates, ensuring tenants are informed about the progress of their requests. It may also allow users to attach photos or descriptions of the issue. Maintenance logs can be stored for future reference and accountability. This system improves response time and helps maintain property quality over time.

Real Time Monitoring - Refers to the continuous tracking of financial transactions as they occur. This includes recording rent payments, expenses, and other financial activities instantly within the system. It enables property owners to access up-to-date information at any given time. With this feature, users can quickly identify issues such as late payments or unexpected costs. Real-time monitoring improves responsiveness and allows immediate action when necessary. It also enhances accuracy by reducing delays in data entry. This feature supports better control over financial operations. Overall, it contributes to more efficient and effective property management.

Payment Tracking - A system function that records and monitors rental payments, including their status such as paid, unpaid, or overdue. It helps property owners keep accurate records of all transactions. The system may include automated reminders for upcoming or missed payments. It can also generate digital receipts for tenants after each transaction. Payment tracking improves transparency between landlords and tenants. It may support multiple payment methods such as cash, bank transfer, or online payments. Historical payment data can be stored for auditing and reference purposes. This feature reduces the risk of missed or unrecorded payments.

Financial Monitoring - The continuous process of tracking rental income, expenses, and other financial activities to ensure proper management of rental properties. It allows property owners to stay aware of their financial status in real time. Regular monitoring helps identify unusual spending or income patterns. It supports better budgeting and financial planning. This process can include tracking utility costs, maintenance expenses, and operational fees. Alerts or notifications may be generated for critical financial updates. It helps prevent financial mismanagement and losses. Consistent monitoring leads to more informed financial decisions.

Financial Reporting - The process of generating structured summaries of financial data such as income, expenses, and overall financial performance for analysis and decision-making. Reports can be generated daily, monthly, or annually. These reports help property owners evaluate profitability and performance. They may include charts, tables, and summaries for easier understanding. Financial reporting supports tax preparation and compliance requirements. It also provides a clear record for audits and reviews. Reports can be exported or shared when needed. This feature improves financial transparency and accountability.

Analytics Dashboard - A visual interface that presents system data through charts, graphs, and summaries to help users easily understand financial performance, bookings, and maintenance activity trends. It provides a centralized view of important system information. Users can quickly identify trends and patterns through visual data representation. Dashboards often include customizable widgets based on user preferences. Real-time updates ensure that the data displayed is always current. It enhances decision-making by presenting complex data in a simple format. Interactive features may allow users to filter or drill down into details. This tool improves overall system usability and efficiency.

Data-Driven Decision Making

Is the process of making informed choices based on accurate and organized data rather than assumptions. In rental property management, it involves analyzing financial records and performance reports. This approach helps property owners identify patterns and trends over time. It allows for better evaluation of income, expenses, and overall profitability. By relying on data, users can reduce risks and avoid costly mistakes. It also supports strategic planning and long-term growth. Data-driven decisions improve efficiency and resource allocation. Ultimately, this approach leads to more effective and successful property management.

Login / Logout

A security feature that allows registered users to securely access the system (login) and safely exit their session (logout). It ensures that only authorized users can access sensitive information. Login typically requires credentials such as a username and password. Some systems may include additional security layers like verification codes. Logout helps prevent unauthorized access, especially on shared devices. Session management may automatically log users out after a period of inactivity. This feature protects user data and system integrity. It is essential for maintaining overall system security.

User Registration (Create Account)

A process that allows new users to create an account by providing required personal information to access the system. It usually includes details such as name, email, and

password. The system may validate the information to ensure accuracy and uniqueness. Email or phone verification may be required to confirm the user's identity. Registration allows users to access personalized features within the system. It also helps maintain a database of users for management and communication purposes. Users can update their account information after registration. This feature serves as the first step in accessing system services.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Introduction

Property management systems have evolved significantly due to advancements in information technology, shifting from manual record-keeping to automated, web-based, and cloud-based solutions. These systems now integrate multiple functions such as financial monitoring, booking management, maintenance tracking, and data analytics to improve operational efficiency.

According to studies, modern property management systems (PMS) are essential tools that help streamline real estate operations by centralizing property data, automating processes, and improving decision-making capabilities through real-time information access (Encaranacion et al., 2025).

This chapter presents related literature and studies that support the development of **PropSight**, focusing on property management systems, financial monitoring, booking systems, maintenance systems, and data-driven decision-making.

2.2 Related Literature

Property Management Systems

Property management systems (PMS) are computerized platforms designed to manage properties, tenants, maintenance activities, and financial transactions in a centralized system. Historically, PMS replaced paper-based systems that were inefficient and prone to errors (Property management system, 2024).

Modern PMS platforms now operate using web-based and cloud-based technologies, allowing real-time access to data and improved scalability. These systems are widely used in real estate, hospitality, and rental industries to manage operational processes more efficiently.

Research shows that PMS platforms improve operational efficiency by integrating property listings, tenant management, billing systems, and maintenance requests into a single system

(Encarnacion et al., 2025). This integration reduces redundancy and enhances data accuracy, making it suitable for small to medium-scale property owners.

Financial Monitoring Systems

Financial monitoring is a core component of property management systems, as it ensures proper tracking of income, expenses, and overall financial performance. Studies indicate that manual financial tracking methods are inefficient and prone to errors, especially in multi-tenant environments.

Automated financial monitoring systems allow property owners to record transactions accurately, generate reports, and analyze financial trends. According to Encarnacion et al. (2025), integrating financial monitoring into property management systems significantly improves decision-making and reduces operational errors.

Furthermore, financial systems that provide real-time data allow property owners to monitor rent payments, overdue balances, and expenses more effectively, improving overall financial transparency.

Booking Systems in Rental Management

Booking systems are widely used in digital platforms to manage reservations and availability. In rental property management, booking systems allow tenants to reserve units in advance while helping property owners manage occupancy efficiently.

Research shows that manual booking processes often lead to inefficiencies such as double bookings, scheduling conflicts, and inaccurate availability tracking. Web-based booking systems solve these issues by providing real-time updates and centralized reservation management (Encarnacion et al., 2025).

Studies also highlight that booking systems improve tenant experience by providing a structured and transparent reservation process, which is essential for boarding houses and short-term rentals.

Maintenance Management Systems

Maintenance management systems are designed to handle repair requests, facility issues, and property upkeep activities. These systems allow tenants to report issues and enable property owners to track and resolve them efficiently.

According to Encarnacion et al. (2025), integrating maintenance management into property systems improves response time and ensures proper documentation of maintenance activities. This reduces communication gaps between tenants and property owners.

Research further shows that maintenance systems contribute to better property preservation and tenant satisfaction by ensuring timely resolution of issues.

Data-Driven Decision Making

Data-driven decision making refers to the process of using structured and analyzed data to guide business decisions. In property management, this includes analyzing rental income, expenses, occupancy rates, and maintenance trends.

Studies show that systems with analytics dashboards help property owners identify patterns such as late payments, increasing expenses, and occupancy fluctuations. This allows for better financial planning and operational improvements (Encarnacion et al., 2025).

Data-driven systems enhance transparency and support strategic decision-making by providing accurate and real-time information.

2.3 Related Studies

Recent research emphasizes the use of technology and data analytics in enhancing property management operations. Chiwhane et al. (2023) launched EstimaRent, a data-driven rental housing optimization system that aims to help real estate professionals make better decisions. The report noted that monitoring rental market data might assist property managers better understand pricing patterns, tenant demand, and overall property performance. The technology uses data analysis tools to help property stakeholders make more accurate financial decisions and increase the efficiency of rental property management.

Similarly, Dai (2020) presented a Property Management Information System (PMIS) aimed at improving property management through effective data management techniques. The study noted that traditional property management methods frequently struggle to accommodate the growing volume of real estate data. Property managers can save time by creating a centralized information system for storing, processing, and retrieving property data. The study also found

that integrating modern information technologies improves financial monitoring, operational efficiency, and service quality in property management.

2.3.1 Local Studies (Philippines Context)

A study conducted in Siargao Island, Philippines, developed and evaluated a web-based property management system designed for property owners, managers, and tenants. The system included features such as property listing, tenant management, maintenance requests, and financial tracking (Encarnacion et al., 2025).

The study found that integrated property management systems significantly improve efficiency, usability, and data accuracy. However, it also highlighted that many small-scale landlords still rely on manual systems, which results in inefficiencies and inaccurate financial records.

These findings emphasize the need for simplified, accessible systems tailored for small to medium-scale rental property owners in the Philippines.

2.3.2 Foreign Studies

Foreign studies highlight the growing importance of cloud-based property management systems in improving real estate operations. These systems provide centralized access to data, real-time updates, and improved scalability for managing multiple properties.

Research shows that integrated systems combining financial monitoring, booking management, and maintenance tracking significantly improve operational efficiency and tenant satisfaction (Property Management System, 2024).

Additionally, studies emphasize that modern PMS platforms enhance transparency, reduce administrative workload, and support better decision-making through analytics and reporting tools.

2.4 Synthesis of the Study

Based on the reviewed literature and studies, property management systems have evolved into integrated platforms that improve efficiency, accuracy, and decision-making in rental property operations.

However, despite these advancements, there remains a gap in systems designed specifically for small to medium-scale landlords, particularly in the Philippine context. Many existing systems are either too complex, costly, or not fully integrated.

Most systems also separate financial monitoring, booking, and maintenance functions, leading to fragmented data management.

Therefore, the proposed system **PropSight** aims to address this gap by providing a unified web-based platform that integrates financial monitoring, booking management, maintenance tracking, and reporting tools into a single system.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study uses a developmental research design, the most appropriate methodology for studies in information technology education that aim to create, test, and evaluate new software systems intended to solve practical, real-world problems. Developmental research, also known as design and development research, provides a systematic framework for both the creation of technological artifacts and the evaluation of their effectiveness in actual or near-actual use contexts.

The developmental research process in this study begins with a thorough analysis of the current financial management practices of small-scale rental property owners, drawing on the reviewed literature and primary data gathered through user interviews. The findings of this needs analysis directly inform the system's functional requirements, design decisions, and evaluation criteria. Following the design and development phases, the completed system is evaluated by a defined group of users and evaluators.

The choice of developmental research is supported by the nature of the problem PropSight is designed to address. The research problem, that small-scale landlords lack an accessible, purpose-built digital tool for financial monitoring and reporting, is a practical one that calls for a practical solution. Developmental research is the appropriate methodology for such problems because it centers on creating and validating that solution rather than on purely descriptive or explanatory inquiry.

3.2 System Development Methodology

The development of PropSight is guided by the Agile Software Development Methodology. Agile is an iterative and incremental approach to software development that emphasizes collaboration, flexibility, continuous testing, and the progressive refinement of features based on feedback gathered at each stage of the process. Rather than requiring a complete and fixed specification of all requirements before development begins, Agile allows the team to plan, build, test, and review in repeated cycles, with each cycle adding new functionality and refining existing features.

The choice of Agile for PropSight is based on several practical considerations. First, the system's requirements are expected to evolve through user testing and feedback. Second, Agile's emphasis on iterative testing means that defects and design issues can be identified and corrected early. Third, the collaborative structure of Agile supports efficient division of work among the five researchers while maintaining overall project coherence.

Ghimire and Charters (2022) found that agile teams consistently reported better project outcomes in terms of quality, responsiveness to change, and alignment with user needs compared to teams using plan-driven methods. Waja, Shah, and Nanavati (2021) similarly found that agile methodologies were especially effective for projects where requirements were not fully defined at the start, a condition that applies directly to PropSight.

The Agile development process for PropSight is organized into five phases, each described in detail below:

Phase 1: Planning

During the planning phase, the research team identifies and documents the functional and non-functional requirements of PropSight through a combination of literature review findings

and direct interviews with potential end users. The team reviews existing research on rental management systems and financial monitoring tools, identifies the key features needed by small-scale property owners, and creates a product backlog: a prioritized list of all desired system features and functions. The planning phase also covers the selection of the technology stack, the identification of development roles and responsibilities among team members, and the establishment of a project timeline divided into development sprints. A stakeholder analysis is also conducted during this phase to identify all groups who will interact with or be affected by the system, including property owners, tenants, and system administrators, and to understand the specific needs and expectations of each group. The goal is to ensure that the development team has a clear, shared understanding of what PropSight needs to do before construction begins.

Phase 2: System Design

In the system design phase, the research team translates the requirements from planning into a comprehensive set of design artifacts that document the intended structure, behavior, and visual presentation of PropSight. These include the Data Flow Diagram (DFD), which maps how data flows through the system; the Entity Relationship Diagram (ERD), which defines the data entities and their relationships; the Use Case Diagram, which identifies the key interactions between the system and its primary users; and user interface wireframes that guide the visual design of the system's pages and navigation. Database normalization is applied during this phase to ensure that the MySQL schema is organized efficiently and that data redundancy is minimized. The user interface wireframes are developed with reference to the usability principles identified in the literature review, prioritizing clarity, consistency, and ease of navigation for non-technical users.

Phase 3: Development

The development phase is where the system is actually built, with the team constructing components across the three architectural layers in line with the design specifications. Development proceeds in a series of sprints, each targeting a specific functional module or set of features. The first sprints focus on building the system's core infrastructure: the database schema, user authentication, and the basic navigation structure of the interface. Later sprints develop the primary functional modules, including property and unit management, tenant and payment tracking, expense recording, financial report generation, and the analytics dashboard. Each sprint ends with an internal review of completed features, and any identified issues are addressed before the next sprint begins. Version control is used throughout to maintain a complete history of all code changes and to facilitate collaborative development among team members.

Phase 4: Testing

The testing phase is conducted both during and after development. During development, each completed module is subjected to unit testing to verify that individual functions perform correctly. Once multiple modules are complete, integration testing checks that they work together and that data flows correctly between the frontend, backend, and database layers. After all major modules are done, system testing evaluates the full PropSight system against its functional and non-functional requirements under realistic conditions. Finally, user acceptance testing involves representative end users in assessing the system's practical utility, ease of use, and overall quality. Issues identified during UAT are documented, prioritized, and addressed before the final evaluation.

Phase 5: Evaluation

The evaluation phase is the final stage of the Agile development cycle for PropSight. During this phase, the system is formally assessed by a defined group of evaluators, including potential end users and student researchers, against the five criteria of functionality, usability, reliability, security, and efficiency. Evaluators perform a defined set of tasks within the system and provide structured feedback through survey instruments. The results are analyzed to assess the degree to which PropSight achieves its stated objectives and to identify areas for potential improvement.

3.3 System Architecture

PropSight is built on a three-tier architectural model that separates the system into three distinct functional layers: the Presentation Layer, the Application Layer, and the Data Layer. This architecture promotes modularity, maintainability, and scalability by giving each layer a well-defined role that can be updated or expanded without disrupting the others.

3.3.1 Presentation Layer (User Interface)

The Presentation Layer is the part of the system that users directly interact with. It includes all visual elements, forms, navigation menus, dashboards, and data displays that make up the user-facing interface of PropSight.

For the Property Owner (Admin), this includes a financial dashboard showing total monthly income, outstanding balances, and expense summaries; property and unit management screens; a tenant management interface; payment recording and tracking screens; expense entry and categorization forms; financial report generation interfaces; and a maintenance request management screen.

For the tenant (User), the layer provides screens for registering an account, browsing available units, submitting booking requests, checking payment status, and submitting maintenance requests. The Presentation Layer is built using HTML5, CSS3, and JavaScript.

3.3.2 Application Layer (*Backend Logic*)

The Application Layer contains all the business logic, data processing functions, and workflow management mechanisms that drive PropSight's core operations. It sits between the Presentation Layer and the Data Layer, receiving inputs from the interface, processing them according to business rules, executing database operations, and returning results to the interface.

Key functions include: validating and processing booking requests; managing financial transaction records; calculating outstanding tenant balances; routing maintenance requests; generating formatted financial reports; and enforcing role-based access controls. The Application Layer is built in PHP, using prepared statements to protect against SQL injection.

3.3.3 Data Layer (*Database*)

The Data Layer is responsible for the persistent storage, organization, retrieval, and management of all data in PropSight. It maintains the integrity, consistency, and reliability of all financial and operational records.

The Data Layer stores user account information and role assignments; tenant profiles and contact details; property and unit records; rental payment records by tenant, unit, and date; operational expense records by category and property; maintenance request data; generated financial report summaries; and activity logs recording significant user actions.

The system uses MySQL as its relational database management system, providing robust support for complex relational data structures and data integrity enforcement through

foreign key constraints. Supabase is considered as a potential cloud-based deployment option for future versions.

This section presents a detailed explanation of PropSight's system architecture, including the design choices made for each layer, the reasons behind those choices, and the technologies used to support the system's functional and non-functional requirements.

The choice of a three-tier architecture for PropSight is motivated by several well-established software engineering principles. Separation of concerns is the most fundamental of these: by dividing the system into three distinct layers, each with a clearly defined responsibility, the architecture ensures that changes to one layer do not require corresponding changes to the others. For example, if the visual design of the user interface needs to be updated, this can be done by modifying only the Presentation Layer without touching the business logic in the Application Layer or the data schema in the Data Layer. This separation makes the system easier to maintain, test, and extend over time.

Scalability is a second important motivation for the three-tier architecture. If the number of properties or tenants managed through PropSight grows significantly, the Data Layer can be scaled independently, for example by moving to a more powerful database server or migrating to a cloud-based database service, without requiring changes to the Application or Presentation Layers. Similarly, if the Application Layer needs to be updated to support new business logic, this can be done without disrupting the user interface or the database structure.

The Presentation Layer of PropSight is built using a standard web technology stack of HTML, CSS, and JavaScript. HTML provides the semantic markup structure for all system pages, using elements such as forms, tables, navigation menus, and data display containers. CSS is used for all visual styling, including the color scheme, typography, layout, and responsive design breakpoints that allow the interface to adapt to different screen sizes. JavaScript is used

for client-side interactivity, including real-time form validation, dynamic content updates without full page reloads, and interactive dashboard chart rendering.

The Application Layer is built in PHP, which serves as the server-side scripting language that processes all system requests and coordinates between the user interface and the database. PHP was selected for the Application Layer for several reasons. First, it is widely supported by low-cost shared hosting environments, which is important for a system designed to be accessible to small-scale property owners with limited IT infrastructure budgets. Second, it has extensive documentation, a large developer community, and a rich ecosystem of libraries and frameworks, making it a practical choice for a student development team. Third, it integrates seamlessly with MySQL, providing efficient and reliable database connectivity. The Application Layer uses PHP's MySQLi extension for database interactions, implementing prepared statements and input validation to enhance security and help protect against SQL injection attacks.

The Data Layer uses MySQL as the primary relational database management system. The database schema is designed following the principles of relational database normalization, with data organized into related tables that minimize redundancy and ensure data integrity. The main tables in the PropSight database correspond directly to the entities identified in the Entity Relationship Diagram: Users, Tenants, Properties, Units, Payments, Expenses, MaintenanceRequests, FinancialReports, and ActivityLogs. Foreign key constraints are used to enforce referential integrity between related tables, ensuring that records in one table always correspond to valid records in the tables they reference.

The PropSight database schema also includes several design features specifically to support the financial monitoring and reporting functions of the system. Payment records include fields for both the payment amount and the expected amount, allowing the system to

automatically calculate the payment status and outstanding balance for each tenant. Expense records include a category field that allows expenses to be grouped and summarized by type in financial reports. A financial_period field is included in both payment and expense records to allow the system to generate reports for specific time periods without requiring complex date range queries.

3.4 Tools and Technologies Used

The development of PropSight uses a carefully selected stack of tools and technologies that are widely adopted in web development and well-suited to the system's technical requirements. The table below summarizes the primary tools and technologies used:

Category	Tool / Technology	Role in PropSight
Frontend	HTML	Structural markup for all system pages, forms, and interface screens
Frontend	CSS	Visual styling, responsive layout, and dashboard presentation
Frontend	JavaScript	Client-side interactivity, dynamic content updates, and form validation
Backend	PHP	Server-side business logic, database connectivity, session management, and report generation
Database	MySQL	Relational data storage, query processing, and data integrity enforcement
Database (Cloud)	Supabase	Cloud-based PostgreSQL option considered for deployment
IDE	Visual Studio Code	Primary code editing environment with extensions for PHP, HTML, CSS, and JavaScript
Local Server	XAMPP	Local development environment hosting Apache, PHP, and MySQL

3.5 Data Gathering Techniques

The data gathered for this study comes from three primary sources: direct interviews with potential property owner end users, direct observation of their current management practices, and a systematic review of related existing studies. These techniques were chosen to ensure that the system's design is grounded in the actual needs and experiences of its target users.

Semi-structured interviews are conducted with a purposely selected group of small to medium-scale rental property owners who currently manage their properties independently. Participants are selected on the basis that they own and manage at least one rental property with two or more units and do not use dedicated property management software. The interview guide covers the following areas: how the participant currently tracks rental income and expenses; specific challenges they face in managing financial records; what information they find most useful for evaluating property performance; and what features they would expect from a digital management system.

Interviews are conducted in person or via video call, are audio-recorded with participant consent, and are transcribed for analysis. Key themes from the interviews are used to refine the system's functional requirements and prioritize features in the product backlog. Direct observation of how potential users currently manage their financial records is conducted alongside the interviews where possible, supplementing the interview findings with firsthand observation data.

The selection of research participants for this study follows purposive sampling principles, which ensure that the individuals interviewed and observed are those most likely to provide relevant and useful data for the research questions being investigated. Purposive sampling is the appropriate sampling strategy for this study because the research is not

attempting to produce statistically representative findings that can be generalized to the entire population of rental property owners in the Philippines. Rather, it is seeking to understand the experiences and needs of a specific subset of that population: small to medium-scale landlords who manage their properties independently using manual or semi-manual financial management methods.

The criteria for participant selection are designed to ensure relevance and diversity. Relevance is ensured by requiring that all participants are currently managing rental properties and are experiencing the financial management challenges that PropSight is designed to address. Diversity is encouraged by seeking participants who vary in the number of units they manage, the type of properties they own, their geographic location within the Iloilo region, and their level of prior experience with digital management tools. This diversity increases the likelihood that the requirements identified through the interviews and observations will reflect the needs of a broad range of potential PropSight users rather than the specific preferences of a narrow and homogeneous group.

The number of participants selected for the interview and observation phase of the study is guided by the concept of information saturation, the point at which additional interviews no longer reveal substantively new themes or requirements. In qualitative research, information saturation is typically reached after a relatively small number of in-depth interviews, often between five and fifteen participants, depending on the homogeneity of the participant group and the complexity of the research questions. For this study, the research team plans to conduct interviews with at least five to eight property owners, with additional interviews conducted as needed until no new functional requirements or design insights are emerging from the data.

The analysis of interview and observation data follows a thematic analysis process adapted from the qualitative research methodology described by Braun and Clarke (2006) and widely used in information systems research. Thematic analysis involves reading and re-reading the interview transcripts and observation notes, identifying recurring ideas and patterns, grouping these into themes, and using the themes to develop a structured understanding of the research problem and its implications for system design. For this study, the thematic analysis focuses specifically on identifying functional requirements (what the system needs to do), usability requirements (how the system needs to behave to be usable by non-technical users), and priority requirements (which features are most urgently needed versus which are secondary or aspirational).

The output of the thematic analysis is a structured requirements document that lists all identified functional and non-functional requirements, organized by priority and mapped to specific system modules. This requirements document forms the basis for the PropSight product backlog, which guides the Agile development process. The requirements document is reviewed and validated by the research team before the start of the first development sprint, and is updated as new requirements are identified during user testing in later phases of the study.

An important methodological consideration in this study is the management of the relationship between the researchers as system developers and the research participants as potential end users. There is a risk that participants may feel pressure to provide positive feedback about the system out of a desire to please the researchers, particularly if the interview and testing sessions are conducted by team members who are also the developers. To mitigate this risk, the research team separates the roles of interviewer and developer where possible, and designs the evaluation questionnaire to include both positively and negatively worded items to reduce acquiescence bias. Participants are also reminded at the beginning of each session

that honest, critical feedback is more valuable to the research than positive feedback, and that there are no right or wrong answers.

The ethical dimensions of the research are also carefully considered. All participants provide informed consent before taking part in interviews, observations, or testing sessions. Consent forms explain the purpose of the research, the use of audio recordings, the confidentiality of responses, and the right to withdraw at any time without penalty. No personal financial information of participants or their tenants is collected, shared, or stored as part of the research process. All data collected is stored securely and is accessible only to members of the research team.

3.6 Testing Procedures

The testing of PropSight uses a multi-level strategy that verifies the system's correctness, reliability, and usability at progressively higher levels of integration and complexity. Each level addresses a different dimension of system quality and provides feedback that informs further development and refinement.

3.6.1 Unit Testing

Unit testing is performed on individual system components and functions to verify that each performs its intended task correctly under a range of input conditions. Unit tests are written and executed by the development team during each sprint, enabling early detection and correction of defects before they affect other parts of the system. For PropSight, unit tests cover functions such as payment recording, expense categorization, report calculation, user authentication, and form validation. Each test case specifies the input, the expected output, and the actual output, and results are documented for review. Any function that fails its unit test is corrected and retested before the sprint review.

3.6.2 Integration Testing

Integration testing examines how different modules of the system interact and verifies that data flows correctly across the boundaries between the frontend interface, the PHP backend logic, and the MySQL database. Integration tests check scenarios such as the correct storage of payments entered through the interface, the accurate retrieval and display of financial report data, the proper enforcement of role-based access controls, and the correct routing of maintenance requests from tenant submission to admin review. Integration testing is conducted after each sprint in which new modules are connected to previously developed components.

3.6.3 System Testing

System testing evaluates the complete, fully integrated PropSight system against its functional and non-functional requirements under conditions that closely simulate real-world use. System tests cover all major use cases, including income recording, expense tracking, report generation, payment monitoring, and maintenance request management, as well as performance and reliability under typical load conditions. Failures identified during system testing are documented in a defect tracking log, prioritized by severity, and corrected before the system proceeds to user acceptance testing.

3.6.4 User Acceptance Testing (UAT)

User acceptance testing is the final stage of the testing process. It involves representative end users, including small-scale property owners and student evaluator groups, in assessing the system's practical usefulness, ease of use, and overall quality. Participants are asked to complete a defined set of representative tasks within PropSight, including recording a rental payment, entering an expense, generating a monthly income report, and submitting a maintenance request. Evaluators observe their interactions and note usability issues or missing

features. After completing the tasks, participants provide structured feedback through the evaluation questionnaire.

3.7 Evaluation Criteria

The evaluation framework for PropSight is grounded in established models for assessing the quality of information systems, adapted to the specific context and objectives of this study.

The primary evaluation model used in this study is the ISO/IEC 25010 Systems and Software Quality Requirements and Evaluation (SQuaRE) standard, which provides a comprehensive framework for assessing software quality across multiple dimensions. From the eight quality characteristics defined in ISO/IEC 25010, five were selected as the most relevant to PropSight's objectives and target user population: functional suitability (corresponding to the functionality criterion in this study), usability, reliability, security, and performance efficiency (corresponding to the efficiency criterion in this study).

The functional suitability characteristic in ISO/IEC 25010 encompasses three sub-characteristics: functional completeness, which asks whether the system provides all the functions needed by users for their specified tasks; functional correctness, which asks whether the system provides the right results with the needed degree of precision; and functional appropriateness, which asks whether the functions facilitate the accomplishment of specified tasks and objectives. All three sub-characteristics are assessed in PropSight's evaluation through the user acceptance testing process, in which evaluators attempt to complete all major system tasks and rate the completeness, correctness, and appropriateness of the results.

The usability characteristic in ISO/IEC 25010 encompasses sub-characteristics including learnability, operability, user error protection, and user interface aesthetics. These dimensions are assessed in PropSight's evaluation through a combination of direct observation during user

acceptance testing and structured questionnaire responses that ask evaluators to rate the ease of learning to use the system, the ease of completing tasks without errors, and the overall quality of the interface design.

The evaluation questionnaire administered during user acceptance testing is adapted from the System Usability Scale (SUS), a widely used and validated instrument for assessing software usability developed by Brooke (1996). The SUS consists of ten items rated on a five-point Likert scale, covering dimensions of usability including learnability, efficiency, and overall satisfaction. The SUS score is calculated on a scale of 0 to 100, with scores above 70 generally considered acceptable, scores above 80 considered good, and scores above 90 considered excellent. In addition to the SUS items, the PropSight evaluation questionnaire includes additional items specific to the five evaluation criteria of this study: functionality, reliability, security, and efficiency.

The evaluation results will be analyzed using descriptive statistics, including means, standard deviations, and frequency distributions for each evaluation item and criterion. These statistics will be used to assess the overall quality of PropSight and to identify specific areas of strength and areas requiring further development.

3.7.1 Functionality

Functionality measures whether PropSight delivers all the features specified in the system requirements and performs them correctly under normal operating conditions. Evaluators verify that all major use cases can be completed without errors or system failures.

3.7.2 Usability

Usability measures how easy it is for target users to learn to use PropSight, navigate its interface, and complete their intended tasks without significant difficulty. Key dimensions include

the clarity of the interface design, the consistency of navigation patterns, the helpfulness of feedback messages, and overall user satisfaction.

3.7.3 Reliability

Reliability measures the consistency and dependability of PropSight in performing its functions correctly and without failure. A reliable system produces consistent results for identical inputs, handles errors gracefully, and maintains the integrity of financial records across all transactions.

3.7.4 Security

Security evaluates the robustness of the mechanisms within PropSight that protect sensitive financial data from unauthorized access, modification, or disclosure, including user authentication strength, role-based access control effectiveness, and database integrity.

3.7.5 Efficiency

Efficiency measures whether PropSight allows property owners to complete their financial management tasks with less time and effort compared to their current methods, through automation of calculations and report compilation, and quick access to key information through the dashboard.

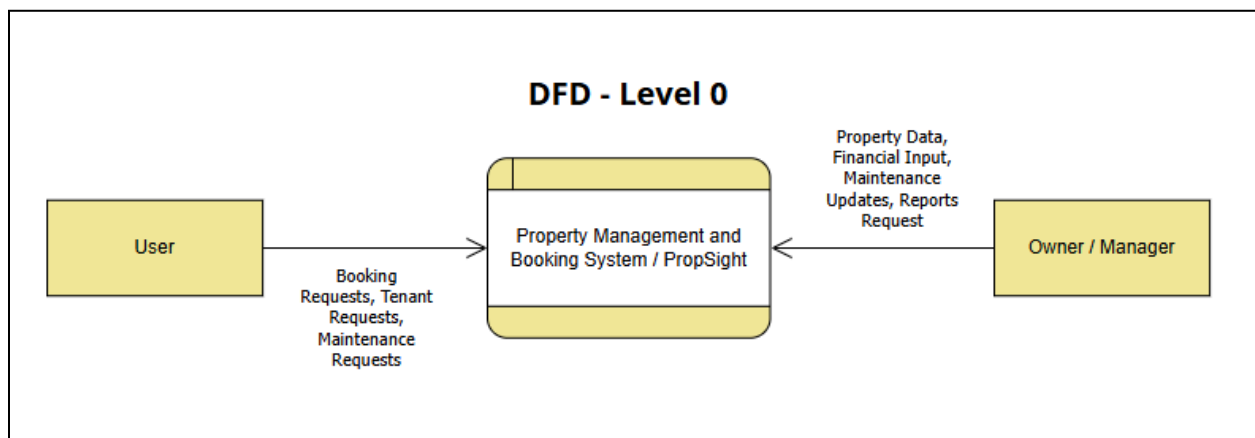
SUPPORTING DIAGRAMS

Data Flow Diagram (DFD – Level 0)

The Level 0 DFD provides a high-level view of how PropSight interacts with its two main external entities: the Property Owner (Admin) and the Tenant (User). At this level, the entire system is represented as a single process, and the diagram shows the types of data flowing between the external actors and the system.

The Property Owner inputs property details, unit information, financial data such as income and expense entries, maintenance updates, and report requests into the system. In return, the system provides financial reports, dashboard analytics, tenant payment statuses, and maintenance request updates. The tenant submits booking requests, maintenance requests, and account registration information. The system responds with booking status confirmations, payment records, unit availability information, and maintenance acknowledgments. All data passes through the central PropSight system, which processes bookings, manages financial records, handles maintenance requests, and stores all data in the database for reporting and analytics.

[DFD Level 0 Diagram - See Attached Figure]



Entity Relationship Diagram (ERD)

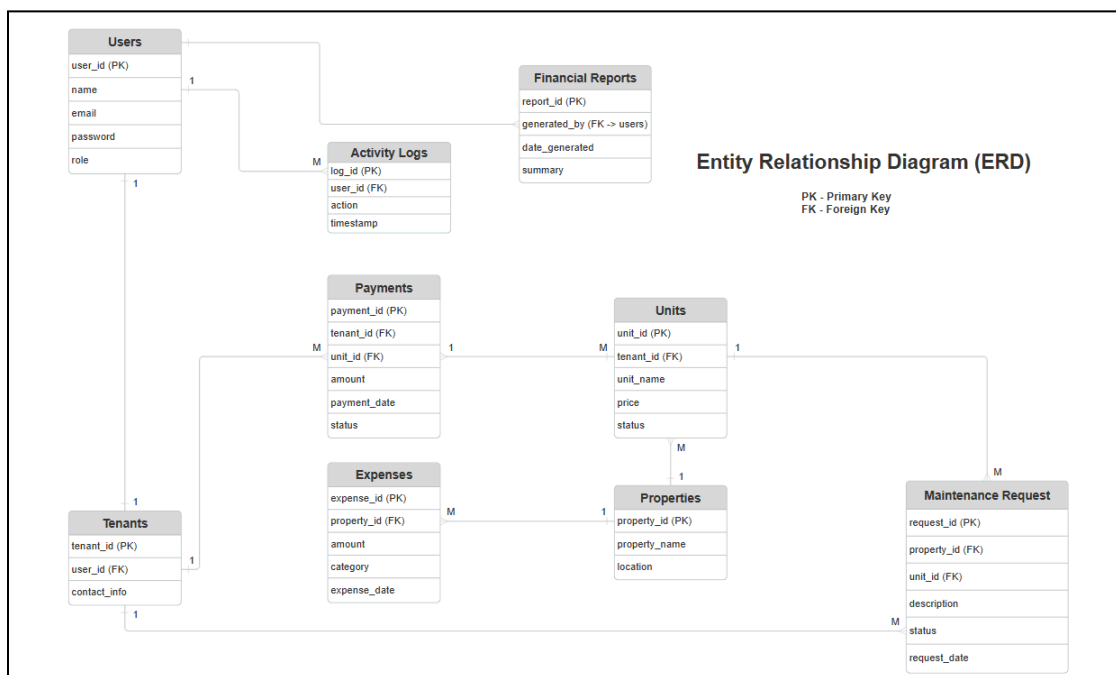
The ERD for PropSight defines the data entities managed by the system and the relationships between them, using standard ERD notation with primary keys (PK) and foreign keys (FK) identified for each entity.

Entity	Attributes
Users	UserID (PK), Name, Email, Password, Role
Tenants	TenantID (PK), UserID (FK), ContactInfo
Properties	PropertyID (PK), PropertyName, Location
Units	UnitID (PK), PropertyID (FK), TenantID (FK), UnitName, Price, Status
Payments	PaymentID (PK), TenantID (FK), UnitID (FK), Amount, PaymentDate, Status
Expenses	ExpenseID (PK), PropertyID (FK), Amount, Category, ExpenseDate
MaintenanceRequests	RequestID (PK), TenantID (FK), UnitID (FK), Description, Status, RequestDate
FinancialReports	ReportID (PK), GeneratedBy (FK - Users), DateGenerated, Summary
ActivityLogs	LogID (PK), UserID (FK), Action, Timestamp

Relationships

- A User can have one Tenant profile (1-to-1 between User and Tenant).
- A User generates many Activity Logs (1-to-Many between User and ActivityLogs).
- A User generates many Financial Reports (1-to-Many between User and FinancialReports).
- A Property contains many Units (1-to-Many between Property and Units).
- A Property has many Expenses (1-to-Many between Property and Expenses).
- A Unit receives many Payments (1-to-Many between Unit and Payments).
- A tenant makes many Payments (1-to-Many between Tenant and Payments).
- A Unit has many Maintenance Requests (1-to-Many between Unit and MaintenanceRequests).
- A Tenant submits many Maintenance Requests (1-to-Many between Tenant and MaintenanceRequests).

[ERD - See Attached Figure]



Use Case Diagram

Use Cases for Property Owner (Admin)

- **Manage Properties and Units:** Add, edit, and monitor rental properties and individual units, including unit status and pricing.
- **Manage Tenant Accounts:** Register new tenants, update tenant profiles, and manage contact information.
- **Manage Bookings:** Review, approve, or decline unit booking requests submitted by tenants.
- **Record Income and Expenses:** Enter rental income payments and operational expenses, categorized by type and property.
- **Monitor Payment Statuses:** Track which tenants have paid and flag overdue accounts.
- **Generate Financial Reports:** Produce income statements, expense summaries, cash flow analyses, and outstanding balance reports.
- **View Analytics Dashboard:** Access real-time charts and summary statistics on financial performance.
- **Handle Maintenance Requests:** Review tenant-submitted maintenance requests and update their resolution status.
- **Manage User Accounts:** Create and manage system user accounts, including assigning roles.
- **View Activity Logs:** Access records of significant system actions for audit and accountability purposes.

Use Cases for Tenant (User)

- Register Account: Create a new user account by providing personal details and contact information.
- Log In and Log Out: Securely access the system and exit when done.
- View Available Units: Browse available rental units with details such as name, price, and status.
- Create Booking Request: Submit a request to reserve an available unit for rental.
- View Booking Status: Check the current status of submitted booking requests.
- View Payment Records: Access personal payment history and check outstanding balances.
- Submit Maintenance Request: Report a problem with the rented unit or property.

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